Short questions

|!|BCESB0130000001|!| ★

The diagrams below show two species of flower that are commonly found in Hong Kong.



Species A



Species B

(a) For each species, describe ***one*** observable feature and explain how it facilitates pollination.

(i) species A (2 marks)

(ii) species B (2 marks)

(b) Suggest ***two*** advantages for a plant to carry out sexual reproduction instead of asexual reproduction. (2 marks)

##

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(a) (i) Any ***one set*** of the following: (2)

Observable feature	***Explanation***
- large flower / brightly coloured petals (1)	to attract insects (1)
- anthers located inside the flower (1)	to brush against the insects in order to transfer pollen grains to the insects (1)
- stigma located inside the flower (1)	to receive pollen grains on the insects (1)

(ii) Any ***one set*** of the following: (2)

Observable feature	***Explanation***
- anthers hang outside the flower (1)	to release pollen grains into the air (1)
- stigmas hang outside the flower (1)	to catch pollen grains in the air (1)

(b) Any ***two*** of the following: (2)

- Offspring show genetic variations which may help them adapt to environmental changes. (1)
- Seeds can be dispersed over long distances, reducing overcrowding and competition. (1)
- Lower chance of disease transmission from parent plants to offspring. (1)

(Or other reasonable answers)

(6 marks)

##

	Skill assessed	**Marks**
(a)(i)	Analysing data	2
(a)(ii)	Analysing data	2
(b)	Understanding basic concepts	2

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|!|BCESB0130000002|!| ★

The following photograph shows a flowering plant called *Gladiolus*.



Source: http://commons.wikimedia.org/wiki/File:Rote_Gladiole.JPG

Credit: 4028mdk09

(a) Is the flower of *Gladiolus* an insect-pollinated flower or a wind-pollinated flower? Explain your answer. (2 marks)

(b) Briefly describe how *Gladiolus* can reproduce asexually. (3 marks)

(c) Suggest ***two*** reasons why it is better for *Gladiolus* plants to reproduce sexually by flowers, rather than reproduce asexually. (2 marks)

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##

(a) Insect-pollinated flower (1)

This is because the flowers are large and with brightly coloured petals / the anthers are inside the flowers. (1)

(b) The leaves of the *Gladiolus* make food by photosynthesis. (1)

Some food is passed down to the underground corm. (1)

A new corm is formed on top of the old corm with the use of the food. (1)

(c) Any ***two*** of the following: (2)

- Offspring show genetic variations which may help them adapt to environmental changes. (1)
- Seeds can be dispersed over long distances, reducing overcrowding and competition. (1)
- The chance of disease transmission from parent plants to offspring is lower. (1)

(Or other reasonable answers)

(7 marks)

##

	Skill assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2

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|!|BCESB0130000003|!| ★

The photograph below shows the structure of a flower.



(a) Name structure X. (1 mark)

(b) State the agent of pollination of the flower in the photograph. Give ***one*** piece of evidence to support your deduction. (2 marks)

(c) (i) With reference to the photograph, suggest why this flower rarely self-pollinates. (2 marks)

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(ii) Give ***one*** advantage and ***one*** disadvantage of self-pollination. (2 marks)

##

(a) Anther (1)

(b) Insects (1)

Any ***one*** of the following: (1)

- The petals are large and brightly coloured. (1)
- The petals have insect guides. (1)
- The stigma and anther are located inside the flower. (1)

(c) (i) The anthers are located at positions lower than that of the stigma. (1)

It is difficult for the pollen grains to reach the stigma of the same flower. (1)

(ii) Advantages (Any ***one*** of the following): (1)

- The chance of retaining desirable characteristics of the parent in the offspring is higher. (1)
- There is a greater chance of successful pollination. (1)
- There is less wastage of pollen grains / gametes / energy. (1)

Disadvantage: (1)

- There are less genetic variations produced in the offspring. (1)

(7 marks)

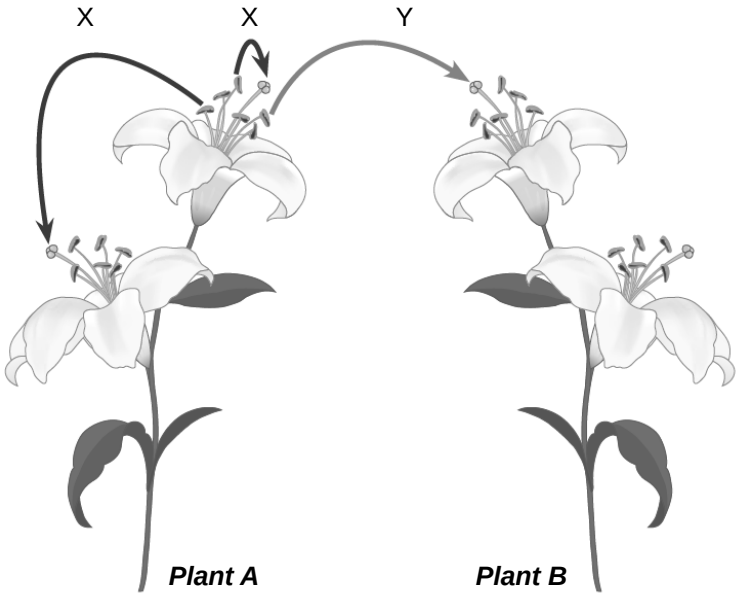
##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Analysing data	2
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	2

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|!|BCESB0130000004|!| ★

The diagram below shows self-pollination and cross-pollination occurring in two flowering plants.



(a) Identify processes X and Y. (2 marks)

(b) What is cross-pollination? (2 marks)

(c) If plant A is *Bauhinia* and plant B is lily, can cross-pollination occur? Explain your answer. (2 marks)

##

(a) X: self-pollination (1)
Y: cross-pollination (1)
(b) Cross-pollination is the transfer of pollen grains from the anther of one plant to the stigma of a flower in another plant of the same species. (1) (1)
(c) No (1)
It is because they do not belong to the same species. (1)

(6 marks)

##

	Skill assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	2
(c)	Applying concepts	2

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|!|BCESB0130000005|!| ★

The following photomicrograph shows some structures produced by the anther of a flower.



(a) Are these structures from an insect-pollinated or a wind-pollinated flower? Explain your answer. (3 marks)

(b) Describe what happens when these structures land on a stigma before fertilization occurs. (4 marks)

##

(a) Insect-pollinated flower (1)
They are rough (1)
to attach easily on insects. (1)

(b) The stigma secretes a sugary solution (1)
which stimulates the pollen grain to develop a pollen tube. (1)
The pollen tube secretes enzymes to digest the tissues of the carpel and grows down through the style and the ovary. (1)
The pollen tube then enters the micropyle and releases the male gamete. (1)

(7 marks)

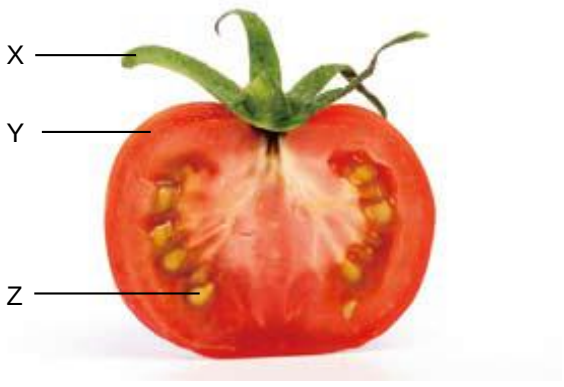
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##

	Skill assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	4

|!|BCESB0130000006|!| ★

The following photograph shows the section of a tomato fruit.



(a) State the external agent required for the seed dispersal of tomato plants. (1 mark)

(b) State the floral structures from which X, Y and Z are developed. (3 marks)

(c) State ***two*** significances of seed dispersal. (2 marks)

##

(a) Animals (1)

(b) X: sepals (1)

Y: ovary wall (1)

Z: ovule (1)

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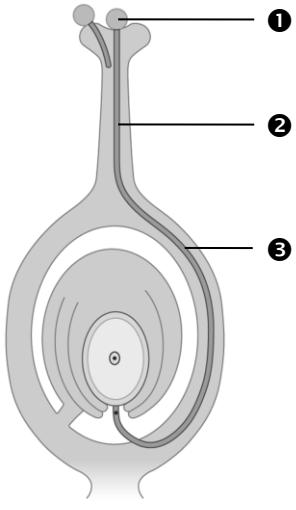
(c) Reduces overcrowding / competition between offspring and the parent (1)
Allow the offspring to colonize new areas (1)
(Or other reasonable answers)

(6 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Analysing data	3
(c)	Understanding basic concepts	2

|!|BCESC0130000007|!|
The diagram below shows the process of fertilization in a flowering plant.



Fill in the blanks with suitable words to explain the process of fertilization. (8 marks)

❶ A pollen grain lands on the (a)_____, germinates and develops a (b)_____.

❷ The (c)_____ secretes (d)_____ to digest the tissues of the carpel.

❸ The (e)_____ grows down through the style and the (f)_____ to carry the male gamete to the ovule through the (g)_____. The male gamete then fuses with the female gamete to form a (h)_____.

##

(a) stigma (1)
(b) pollen tube (1)
(c) pollen tube (1)
(d) enzymes (1)
(e) pollen tube (1)
(f) ovary (1)
(g) micropyle (1)
(h) zygote (1)

(8 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1
(g)	Understanding basic concepts	1
(h)	Understanding basic concepts	1

|!|BCESC0130000008|!|

Complete the table below to compare asexual and sexual reproduction. (4 marks)

	Asexual reproduction	Sexual reproduction
(a) Fusion of gametes		
(b) Number of parents		
(c) Type of cell division involved		
(d) Genetic make-up of offspring		

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##

(a) No; Yes (1)

(b) One; Usually two (1)

(c) Mitotic cell division; Meiotic cell division and mitotic cell division (1)

(d) Identical to the parent; Different from parents (1)

(4 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

|!|BCESC0130000009|!|

For each of the floral parts listed in column I, select from column II one phrase that matches it. Put the appropriate letter in the space provided. (4 marks)

Column I		***Column II***
Anther	(a)_____	A. Fertilization occurs
Stigma	(b)_____	B. Secretes a sugary solution
Filament	(c)_____	C. Supports the stigma
Ovule	(d)_____	D. Supports the anther
		E. Meiosis occurs to produce pollen grains

##

(a) E (1)

(b) B (1)

(c) D (1)

(d) A (1)

(4 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

|!|BCESC0130000010|!|

For each of the structures listed in Column 1, select from Column 2 one phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column I		***Column II***
Ovule	(a)_____	A. develops into the embryo after fertilization
Integument	(b)_____	B. develops into the fruit wall after fertilization
Ovary wall	(c)_____	C. develops into the seed after fertilization
		D. develops into the seed coat after fertilization

##

(a) C (1)
(b) D (1)
(c) B (1)

(3 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESC0130000011|!|

Complete the paragraph below with suitable words. (5 marks)

(a)_____ is the transfer of pollen grains from the anther to the (b)_____. This process requires the help of (c)_____ such as insects and wind. When pollen grains land on the stigma of the same plant, this is called (d)_____. Conversely, when pollen grains land on the stigma of a different plant of the same species, it is called (e)_____.

##

(a) Pollination (1)
(b) stigma (1)
(c) external agents (1)
(d) self-pollination (1)
(e) cross-pollination (1)

(5 marks)

##

	Skill assessed	Mark
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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